

DMS Sizing & Architecture Guide

Instance Selection, Workload Sizing, Architecture Patterns & Best Practices

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1 · Instance Type Selection

Choose the right DMS replication instance family based on workload characteristics



T3 — General Purpose

Burstable CPU. Ideal for development, testing, and small one-off migrations where sustained throughput is not required.



C5/C6i/C7i — Compute Optimized

High clock speed per vCPU. Best when heavy transformations, datatype conversions, or high-throughput parallel load dominate the workload.



R5/R6i/R7i — Memory Optimized

Large RAM per vCPU. Best for large databases, high-volume CDC streams, LOB data, and workloads requiring extensive in-memory buffers.

Family	Instances	vCPU Range	RAM Range	Primary Use Case	Notes
T3	dms.t3.micro → t3.large	2 – 2	1 – 8 GB	Dev / test, small migrations (<10 GB)	Burstable CPU — avoid for sustained production CDC; throttles under load
C5	dms.c5.large → c5.24xlarge	2 – 96	4 – 192 GB	Transformation-heavy loads, schema conversions, high parallelism	Best CPU:RAM ratio when compute is the bottleneck; C6i/C7i are newer equivalents
R5	dms.r5.large → r5.24xlarge	2 – 96	16 – 768 GB	Large DBs, high-TPS CDC, LOB columns, Full Load + CDC (default recommended)	High memory:vCPU ratio; R6i/R7i are newer equivalents with better price-perf
R5 (CDC)	dms.r5.xlarge → r5.4xlarge	4 – 16	32 – 128 GB	Low-latency ongoing CDC; high-TPS CDC streams	Three in-memory buffers (incoming, outgoing, sorter) consume significant RAM

2 · Sizing by Workload

Map your migration size and pattern to a recommended replication instance configuration

Migration Size	Task Type	Recommended Instance	vCPU	RAM	Key Settings
< 10 GB	Full Load	<code>dms.t3.large</code>	2	8 GB	Default settings; no special tuning needed
10 – 100 GB	Full Load	<code>dms.r5.large</code>	2	16 GB	<code>CreatePkAfterFullLoad=true</code> ; drop indexes pre-load
100 GB – 1 TB	Full Load	<code>dms.r5.2xlarge</code>	8	64 GB	Parallel load (partition/range), drop indexes; <code>MaxFullLoadSubTasks=16</code>
> 1 TB	Full Load	<code>dms.r5.4xlarge+</code>	16+	128 GB+	Multi-instance architecture; <code>BatchApplyEnabled=true</code> ; split tables across tasks
Any size	Full Load + CDC	<code>dms.r5.2xlarge+</code>	8+	64 GB+	Extra buffer for cached changes; disable target Multi-AZ during load phase
Low-latency	CDC Only	<code>dms.r5.xlarge</code>	4	32 GB	Monitor <code>CDCLatencyTarget</code> ; <code>BatchApplyEnabled=true</code>
High-throughput	CDC Only	<code>dms.r5.2xlarge+</code>	8+	64 GB+	<code>BatchApplyEnabled=true</code> ; separate heavy-DML tables into dedicated tasks
Dev / Test	Any	<code>dms.t3.medium</code>	2	4 GB	Cost-optimized; do not use for production sustained workloads

 **AWS Instance Estimator:** In the DMS console, when creating a replication instance, use "**Estimate instance class and storage**" — it recommends instance class, storage, and cost based on your migration type, LOB handling, and task parameters.

3 • Key Sizing Principles

Understand the three resource constraints that drive DMS instance selection

1 Memory is the primary constraint

DMS buffers incoming changes in RAM during full load, holding them until the load phase completes and CDC catch-up can begin. Three distinct memory buffers are in use simultaneously: the *incoming stream buffer* (reading from source), the *outgoing stream buffer* (writing to target), and the *sorter/cache layer* (ordering transactions). LOB columns amplify memory pressure significantly — each LOB row may require additional row-by-row lookups and staging. For high-TPS CDC workloads, favor R-family (memory-optimized) instances. Monitor `MemoryUsage` and swap utilization via CloudWatch.

2 CPU drives parallel load performance

`MaxFullLoadSubTasks` (default 8, maximum 49) controls how many tables load in parallel. Each subtask consumes a thread and a portion of vCPU. To maximize throughput, increase `MaxFullLoadSubTasks` to match available vCPUs — an r5.2xlarge with 8 vCPUs can efficiently run 8–16 subtasks. For intra-table parallelism on large tables, configure `ParallelLoadThreads` with partition-based or range-based segmentation. C-family instances provide the best CPU:RAM ratio when the bottleneck is compute rather than memory (e.g., complex transformations, datatype conversions).

3 Storage acts as a safety net

DMS uses local SSD storage on the replication instance as overflow when in-memory buffers are exhausted. Allocate a minimum of

100 GB

for small migrations and

200–500 GB

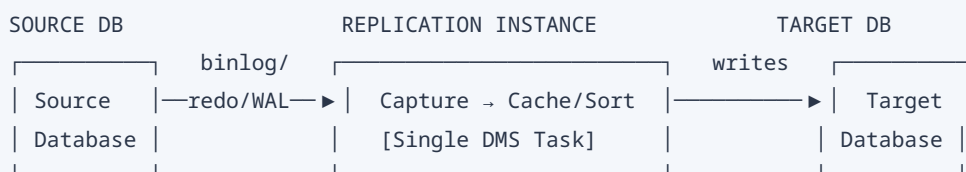
for large migrations with LOB overflow. Storage is also used for task logs. High swap usage on the replication instance is a leading indicator of memory exhaustion — provision more RAM or split tasks before swap causes task failure. Always monitor disk utilization via the DMS console or CloudWatch.

4 • Architecture Patterns

Choose the right deployment topology based on migration scale and table complexity

Pattern A — Single Task (Simple)

► SINGLE TASK ARCHITECTURE



All tables processed by a single task on one replication instance.

✓ BEST FOR

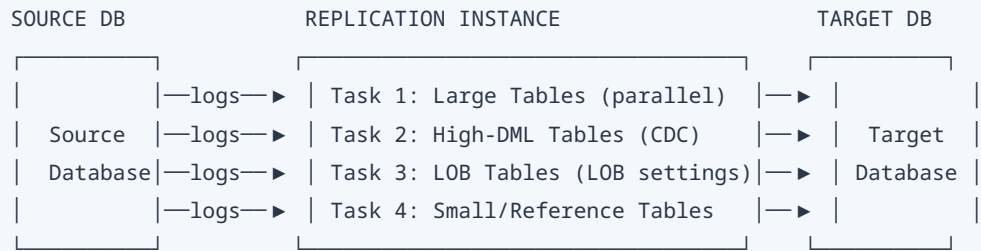
- Small databases (<50 tables, <100 GB)
- Homogeneous migrations (minimal transformation)
- Proof-of-concept and dev/test workloads

⚠ RISK

- One slow table blocks CDC catch-up for all tables
- Single point of resource contention
- LOB tables can starve non-LOB tables

Pattern B — Multi-Task (Recommended for Medium/Large)

► MULTI-TASK ARCHITECTURE — SINGLE REPLICATION INSTANCE

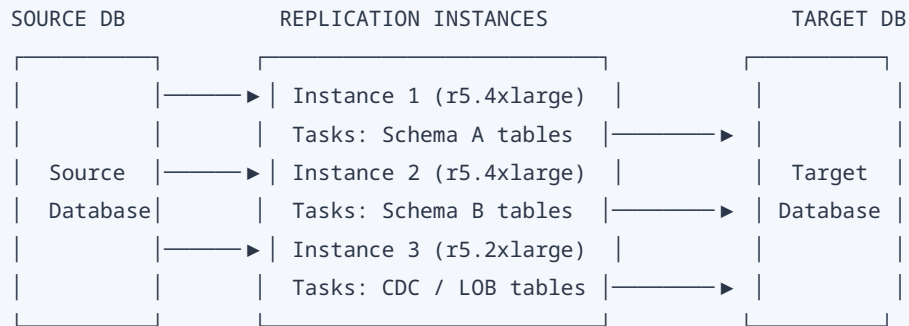


Multiple tasks on a single instance allow independent tuning per table group. Each task can have different LOB mode, commit rate, and parallel load settings.

Task Group	Tables Included	Key Settings
Task 1 — Large Tables	Tables >10 GB each	Partition/range parallel load; <code>ParallelLoadThreads=4+</code>
Task 2 — High-DML Tables	Tables with frequent INSERT/UPDATE/DELETE	<code>BatchApplyEnabled=true</code> ; monitor <code>CDCLatencyTarget</code>
Task 3 — LOB Tables	Tables with BLOB/CLOB/TEXT columns	Limited LOB mode; set <code>LobMaxSize</code> ; avoid Full LOB mode
Task 4 — Reference Tables	Small lookup/reference tables	Default settings; minimal tuning required

Pattern C — Multi-Instance (Enterprise Scale)

► MULTI-INSTANCE ARCHITECTURE — >1 TB MIGRATIONS



Each replication instance handles a distinct subset of tables. Provides fault isolation, independent scaling, and maximum aggregate throughput.

✓ BENEFITS

- Fault isolation — one failed instance doesn't affect others
- Independent scaling per workload type
- Maximum aggregate throughput for >1 TB datasets
- Separate Multi-AZ settings per instance

⚠ CONSIDERATIONS

- Higher cost (multiple instance hours billed)
- More complex orchestration and monitoring
- Requires careful table partitioning to avoid cross-dependency issues

5 • Full Load + CDC — Phased Architecture

The recommended three-phase approach for live database migrations with zero data loss

Phase 1 — Full Load

- Instance: `r5.2xlarge+`
- Drop indexes & FKs on target
- Disable Multi-AZ on target RDS
- Disable automated backups on target
- Set `CreatePkAfterFullLoad=true`
- Use Limited/Inline LOB mode
- Enable parallel load for large tables
- Start during low-traffic window

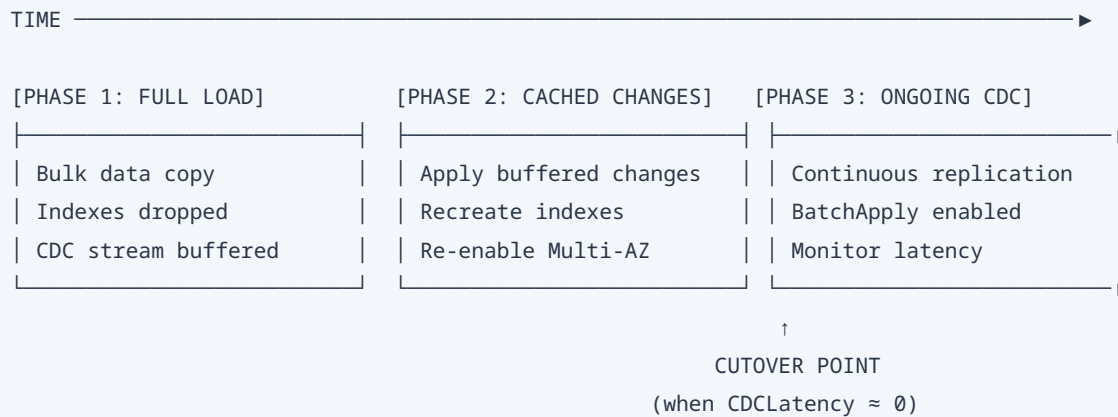
Phase 2 — Cached Changes Apply

- DMS applies changes buffered during full load
- Use `"Stop after applying cached changes"` setting to pause
- Recreate indexes & foreign keys on target
- Re-enable Multi-AZ on target RDS
- Re-enable automated backups
- Validate row counts and checksums
- Prepare cutover runbook

Phase 3 — Ongoing CDC

- Resume task from paused state
- Enable `BatchApplyEnabled=true`
- Monitor `CDCLatencySource` & `CDCLatencyTarget`
- Optionally downsize replication instance (in-place resize supported)
- Execute cutover when latency approaches zero
- Decommission source after validation

► FULL LOAD + CDC TIMELINE



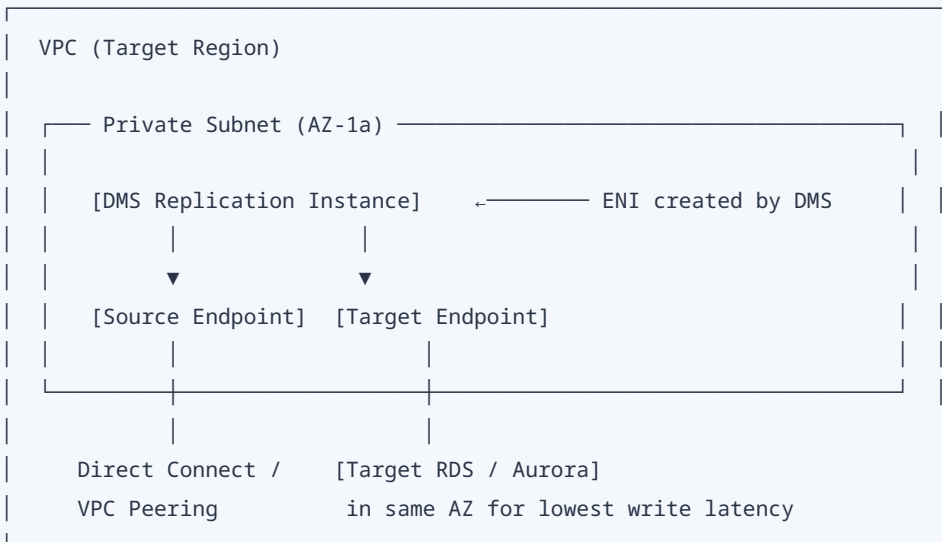
6 · Network & VPC Architecture

Connectivity patterns and placement recommendations for optimal throughput and security

Connectivity Pattern	Use Case	Recommended Setup	Latency Profile
Same VPC — Same AZ	Source and target both in AWS, same VPC	Co-locate replication instance in same AZ as target for write optimization; uses direct routing	LOWEST
Same VPC — Cross-AZ	Source and target in same VPC, different AZs	Acceptable; cross-AZ data transfer charges apply; replication instance placed in target AZ	LOW
Cross-VPC (Same Region)	Source and target in different VPCs	VPC Peering or Transit Gateway; replication instance requires route tables to both VPCs	LOW
Cross-Region	Migrating across AWS regions	Replication instance in target region; source endpoint uses public or VPN connectivity; higher latency expected	MEDIUM
On-Premises Source	Source DB is on-premises or in a colo DC	AWS Direct Connect (preferred for large migrations); Site-to-Site VPN as fallback; DMS instance in target VPC	MEDIUM
Public Internet	Non-sensitive small migrations only	SSL/TLS required on endpoints; not recommended for large datasets or production CDC	VARIABLE

VPC Placement Best Practices

► RECOMMENDED VPC TOPOLOGY (CO-LOCATED WITH TARGET)



DMS creates an ENI (Elastic Network Interface) within your VPC subnet. Place the replication instance in the same AZ as the target database to minimize cross-AZ write latency and data transfer costs.

Security Group Requirements

- Replication instance SG: allow outbound to source DB port (e.g., 3306, 5432, 1521)
- Replication instance SG: allow outbound to target DB port
- Source DB SG: allow inbound from replication instance SG
- Target DB SG: allow inbound from replication instance SG
- No inbound rules required on replication instance (it initiates connections)

Network Throughput Tips

- Larger instance types have higher network bandwidth limits
- For >1 TB migrations, use `r5.4xlarge+` to avoid NIC saturation
- Direct Connect bandwidth should match peak DMS throughput (often 100–500 Mbps for large loads)
- Enable Enhanced VPC Routing for Redshift targets to keep traffic within AWS backbone

7 • Multi-AZ Considerations

When to enable Multi-AZ on the replication instance and target database

Configuration	Phase	Recommendation	Reason
DMS Replication Instance — Single-AZ	Full Load	RECOMMENDED	Better performance — no standby sync overhead; a failed instance can be restarted from last checkpoint
DMS Replication Instance — Multi-AZ	Production CDC	RECOMMENDED	Automatic failover in ~1–2 minutes; minimizes replication downtime if the primary instance fails
Target RDS — Multi-AZ Disabled	Full Load phase only	RECOMMENDED	Eliminates synchronous standby replication overhead; can double write throughput during bulk load
Target RDS — Multi-AZ Enabled	Post Full Load + CDC	REQUIRED	Production resilience requirement; re-enable before cutover
Target RDS — Automated Backups Disabled	Full Load phase only	RECOMMENDED	Reduces I/O contention during bulk load; re-enable immediately after full load completes
DMS Replication Instance — Resize	Post Full Load, CDC-only phase	CONSIDER	DMS supports in-place instance class changes; downsize after full load to reduce ongoing CDC cost

 **Failover Behavior:** When DMS Multi-AZ is enabled and the primary replication instance fails, DMS automatically promotes the standby. In-flight transactions may need to be replayed from the last committed checkpoint — ensure your source transaction logs have sufficient retention (minimum 24 hours recommended) to support this replay window.

8 • Common Sizing Mistakes to Avoid

The most frequent configuration errors that lead to performance degradation, task failures, or data loss

✗ Using T3 Instances for Production CDC

T3 instances use burstable CPU credits. Under sustained CDC load, the instance exhausts its CPU credit balance and throttles to baseline performance — causing CDC latency to spiral. Always use R5 or C5 instances for production CDC workloads.

✗ Single Task for All Tables

Placing all tables in a single DMS task means one slow table (e.g., a large LOB table) can block CDC catch-up for all other tables in the task. Split tables by workload type: large tables, high-DML tables, LOB tables, and reference tables should each have dedicated tasks.

✗ Ignoring Swap Usage

High swap usage on the replication instance is the clearest early warning of memory exhaustion. If `SwapUsage` is consistently non-zero, the instance is spilling to disk — task failure risk increases significantly. Right-size the instance or split tasks to reduce per-task memory pressure before it causes a crash.

✗ Not Monitoring CDCLatencyTarget

`CDCLatencyTarget` is the most critical CDC health metric — it measures the delay between when a change is captured from the source and when it is applied to the target. A growing `CDCLatencyTarget` indicates the target apply is falling behind. This is often the first symptom of under-provisioning. Monitor it continuously, not just at cutover.

✗ Keeping Multi-AZ Enabled on Target During Full Load

When Multi-AZ is enabled on an RDS target, every write is synchronously replicated to the standby before acknowledging. During full load, this doubles the effective write I/O and can halve throughput. Temporarily disable Multi-AZ and automated backups on the target during the full load phase, then re-enable before cutover.

✗ Using Full LOB Mode Without Justification

Full LOB mode performs a separate row-by-row lookup for every LOB value — this can reduce full load throughput by 10x or more compared to Limited LOB mode. Use `Limited LOB` mode with an appropriate `LobMaxSize` for the majority of LOB data. Only use Full LOB mode if you have LOB values that exceed the maximum size supported by Limited mode.

✗ Undersizing Storage on the Replication Instance

DMS uses local SSD storage for task logs, cached changes overflow, and LOB staging. Insufficient storage causes task failures mid-migration. Provision at least 100 GB for small migrations and 200–500 GB for large or LOB-heavy workloads. Increase storage proactively rather than reactively — DMS does not support online storage reduction once allocated.

✗ Starting Full Load + CDC During Peak Traffic

Starting a Full Load + CDC task during peak source traffic maximizes the size of the CDC backlog that accumulates during the load phase. A large backlog means Phase 2 (cached changes apply) takes longer, extends the maintenance window, and increases cutover risk. Always start the task during the lowest-traffic period of the week.